Chapter 1
Introduction to Microelectronics

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Introduction to Microelectronics

- Electronics versus Microelectronics
- Examples of Electronic System
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  - ✓ Digital Camera
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  - ✓ Circuits
- Basic Circuit Theorems
  - ✓ KCL
  - ✓ KVL

1.1 Electronics versus Microelectronics

- **Electronics**
  - ✓ Vacuum tubes:

    Used for radio communications and radar systems during WWII (1939–1945).

[1]

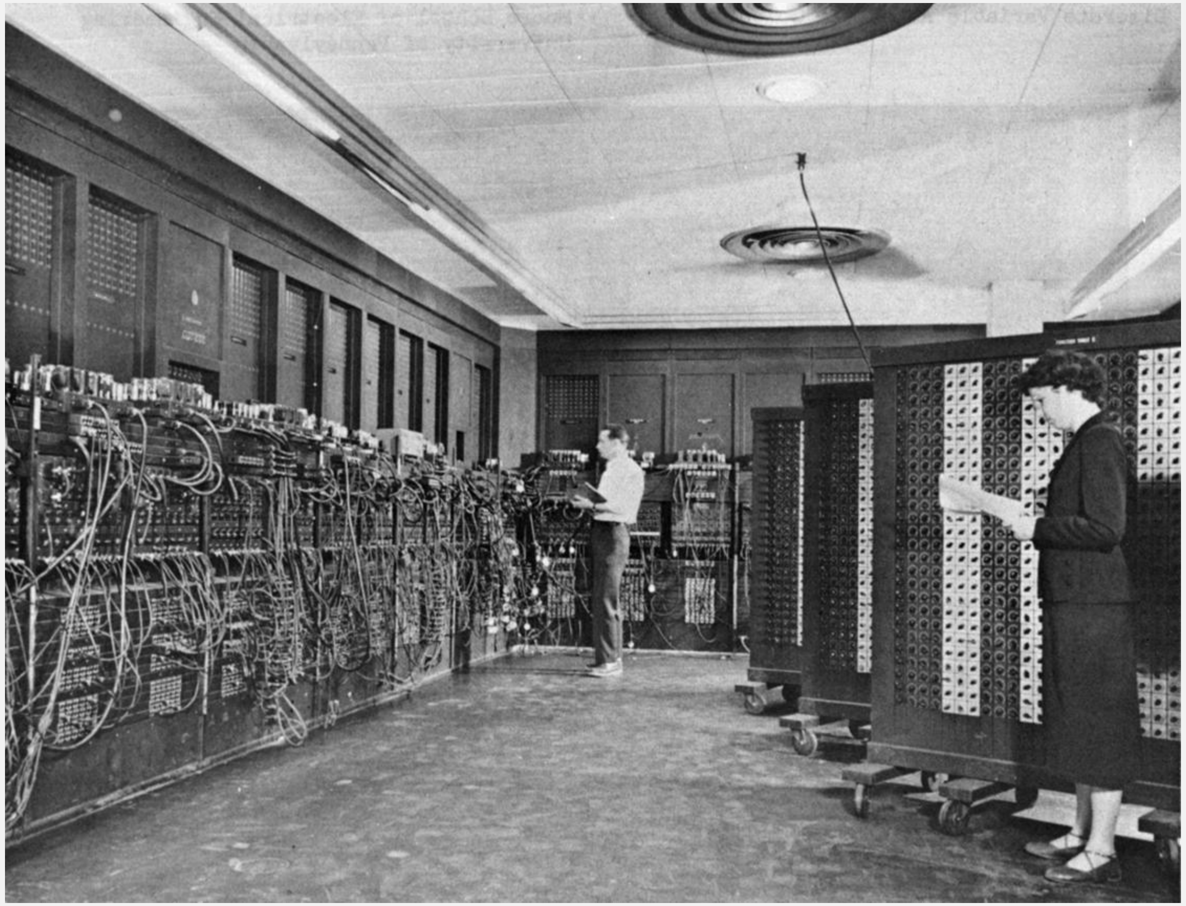
Audio Power Amplifier

[2]

Vacuum tube-based audio power amplifier

The First “Digital” Computer in the World

- **ENIAC** (Electronic Numerical Integrator and Computer)



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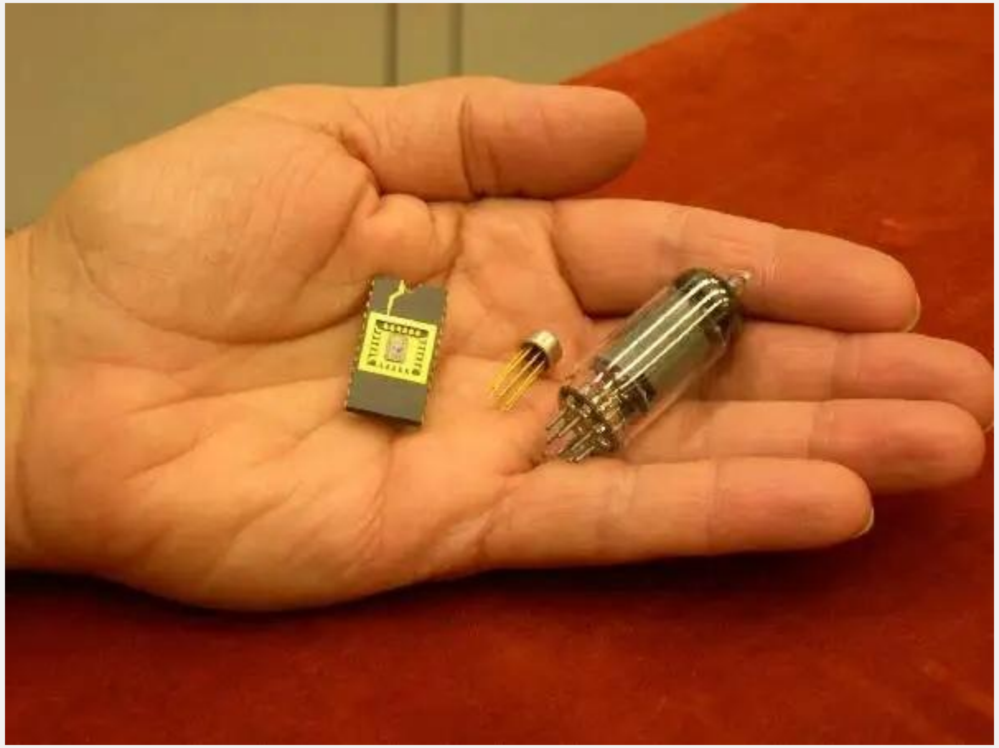
Early vacuum tube computer (ENIAC)

Disadvantage of Vacuum Tubes

✓ Cons (**Problem?**):

- ➢ Finite lifetime
- ➢ The large size of vacuum tubes



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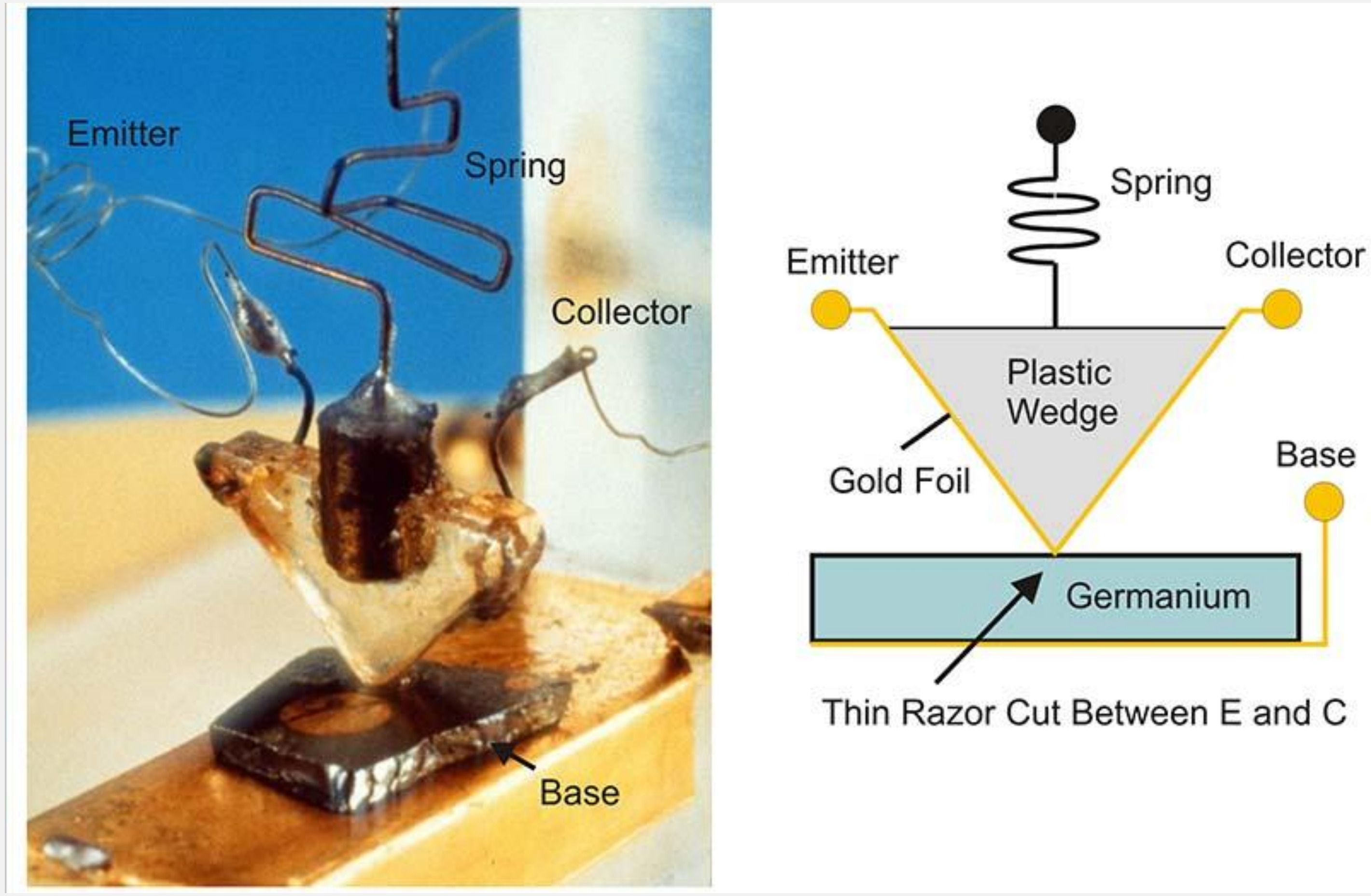
IC, transistor, and vacuum tubes

The First Transistor in the World
The "Big Bang" of Microelectronics

- Point-contact transistor (1947)



[5]

The Inventors of the First Transistor

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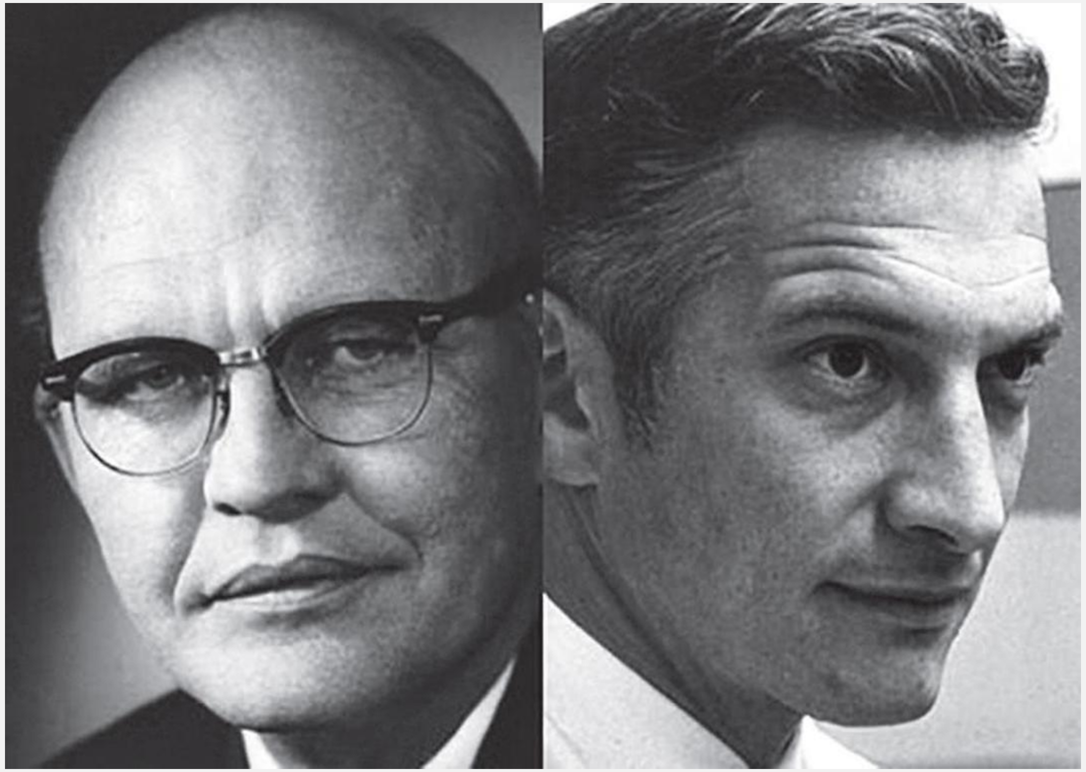
John Bardeen, William Shockley, and Walter Brattain at Bell Labs (1948)

Who invented the first IC?

- **Patent War: Texas Instruments vs. Failchild**

Texas Instruments

(1951~now)



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Jack Kilby vs. Robert Noyce

Fairchild Semiconductor

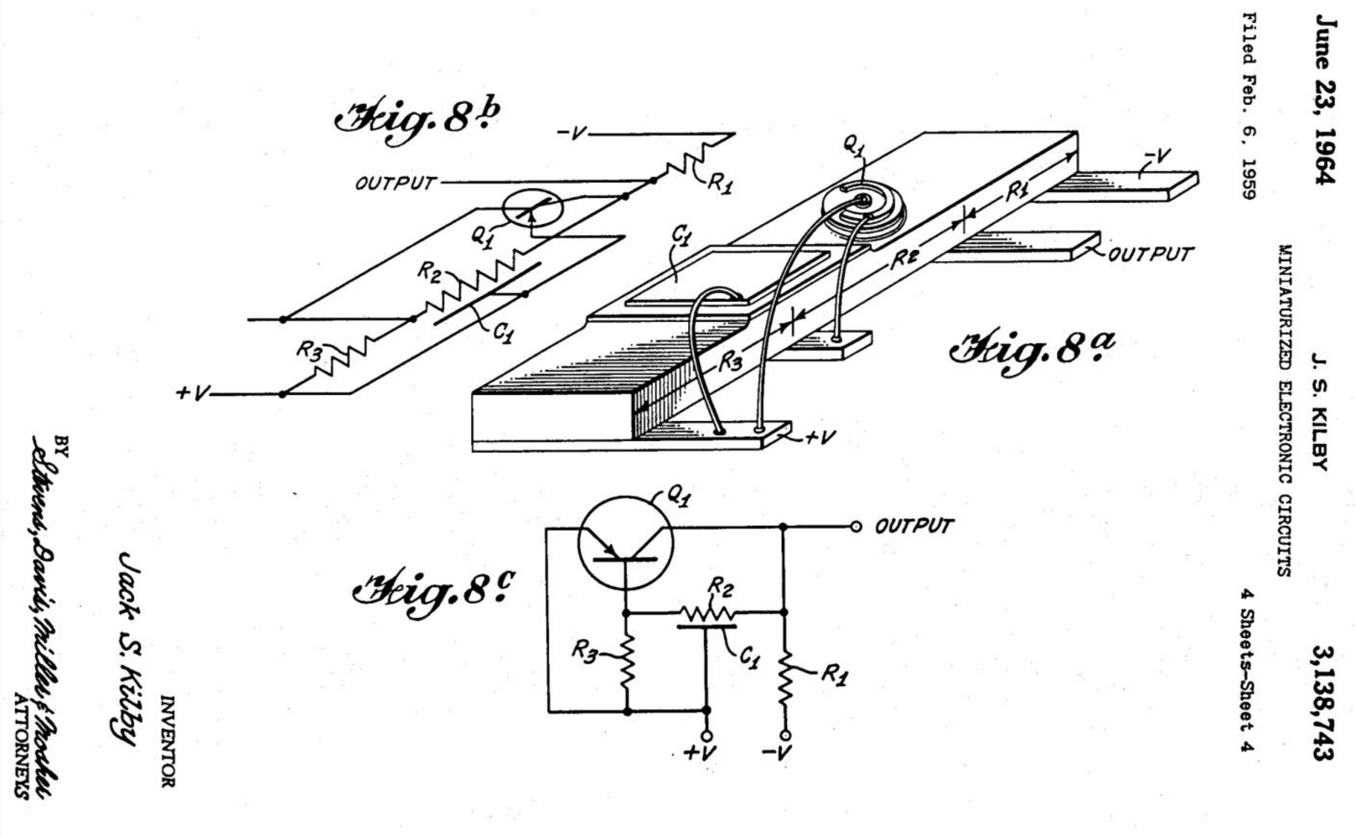
(1957~2016)

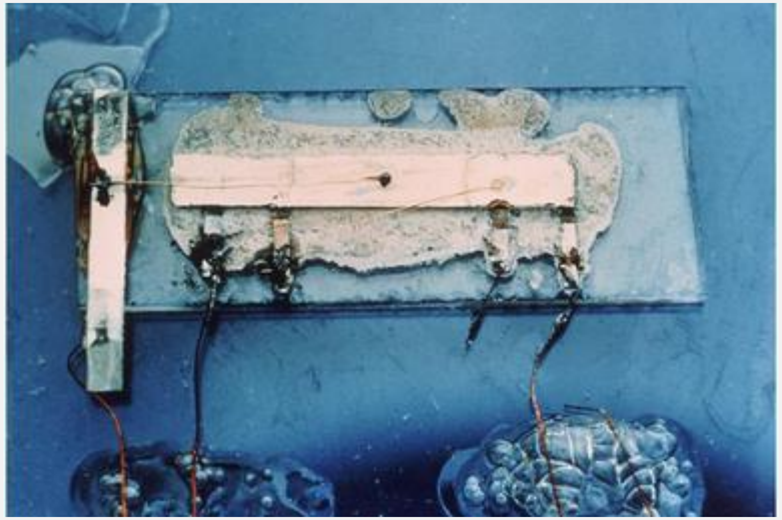
intel

(1968~now)

Jack Kilby's IC (1958)

- Problem: the flying wire

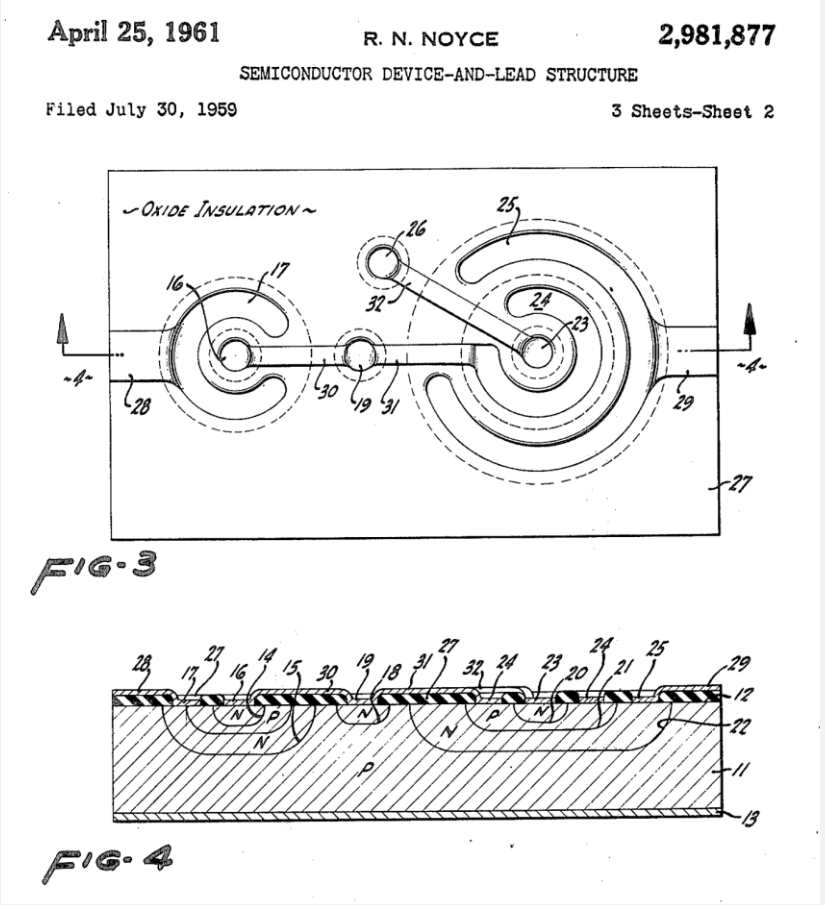


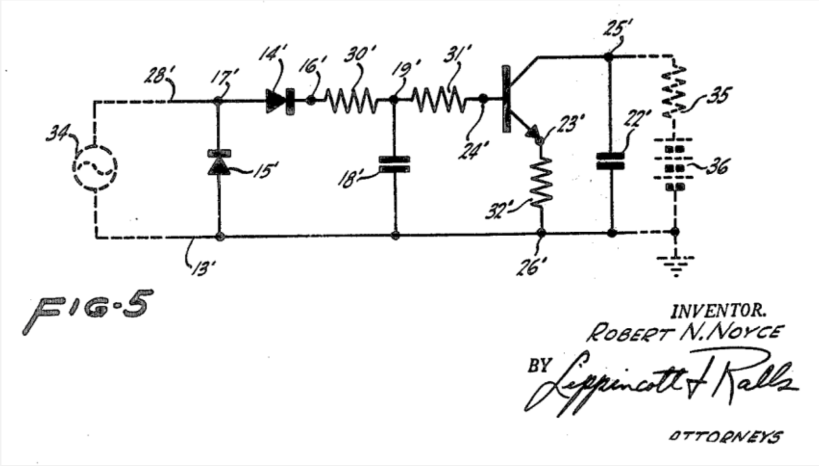

[8]

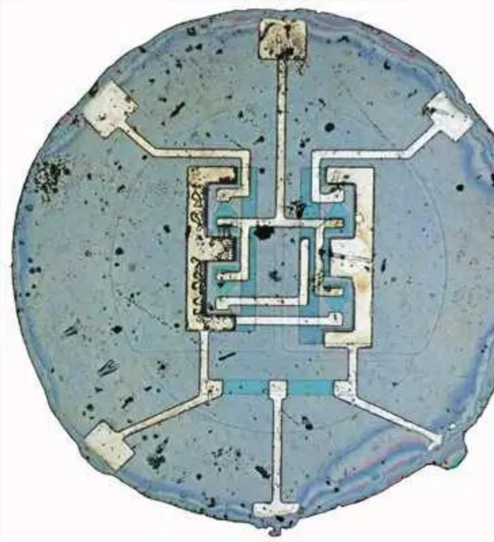
Miniaturized electronic circuits, U.S. Patent 3,138,743 (Filed 1959/2/6, Issued 1964)

Robert Noyce’s IC (1959)

- Pros: the origin of modern IC process (the planar process)





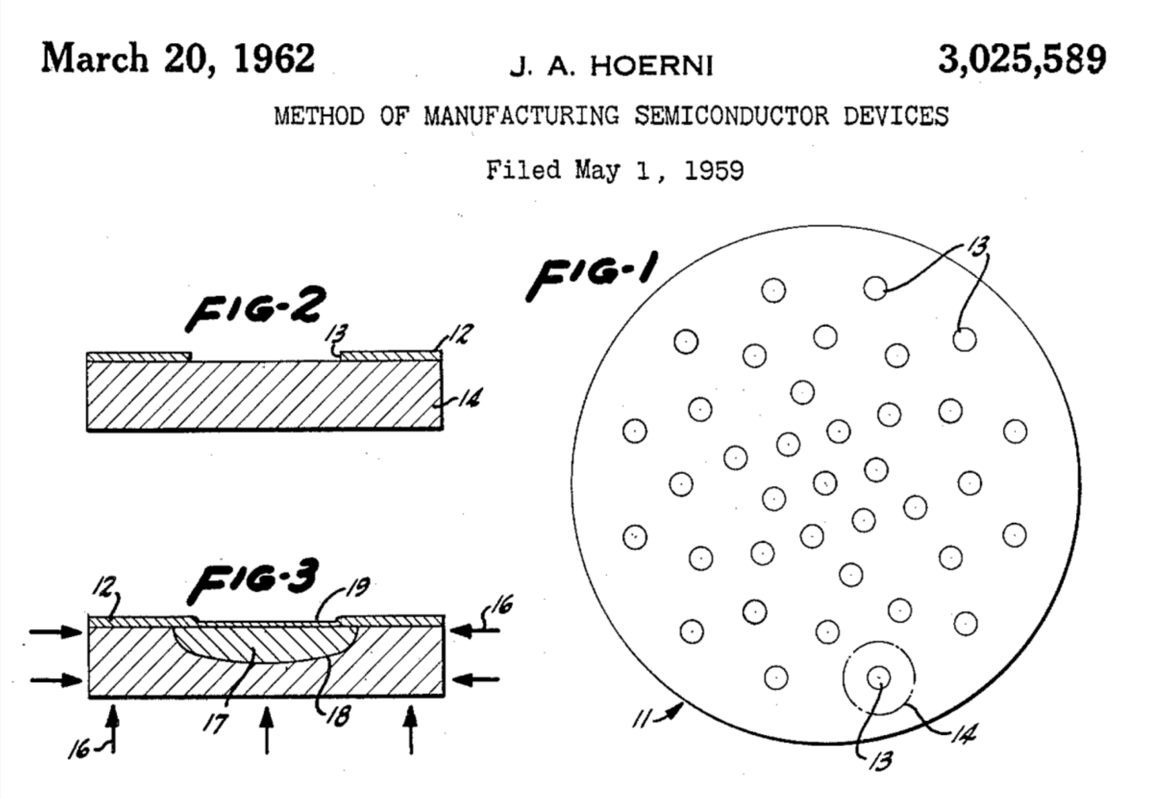


[9]

Semiconductor device-and-lead structure, U.S. Patent 2,981,877 (Filed 1959/7/30, Issued 1961)

Inventor of the "Planar" Process for IC

- Jean Hoerni

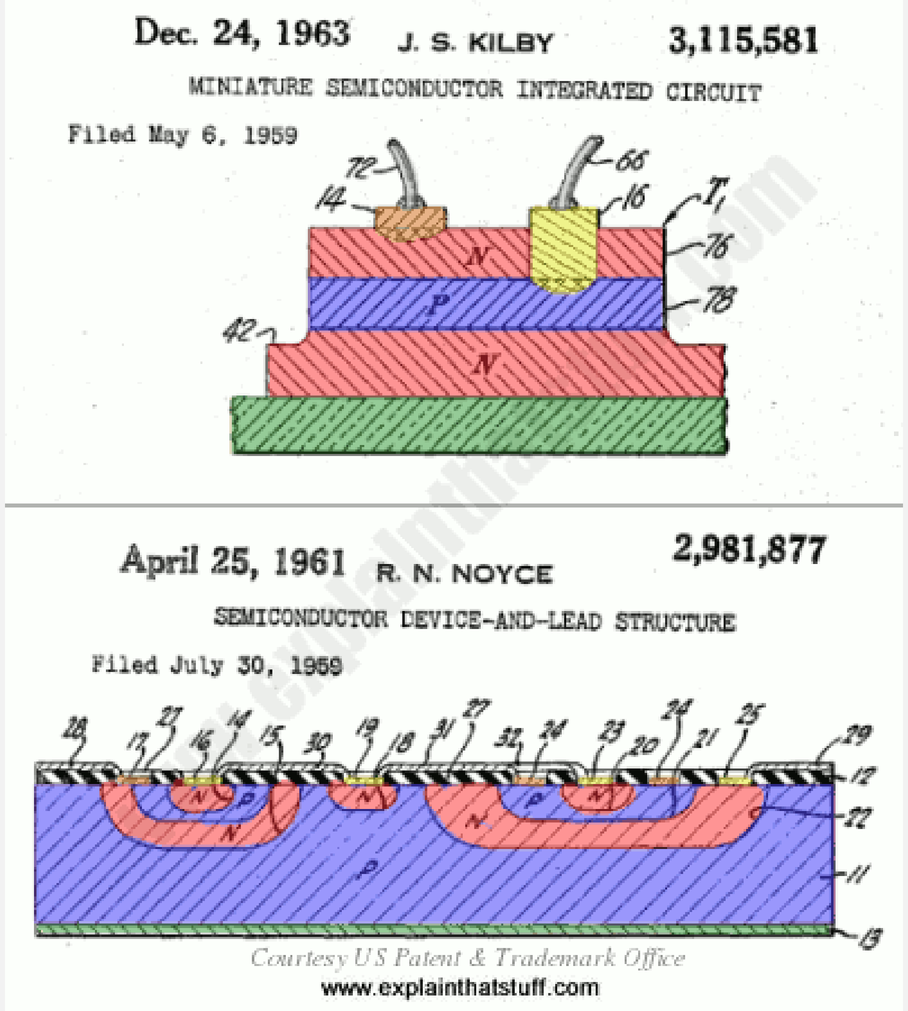


Method of manufacturing semiconductor devices, U.S. Patent 3,025,589 (Filed 1959/5/1, Issued 1962)

[10]

Jean Hoerni

The Flying Wire vs. the Planar Process



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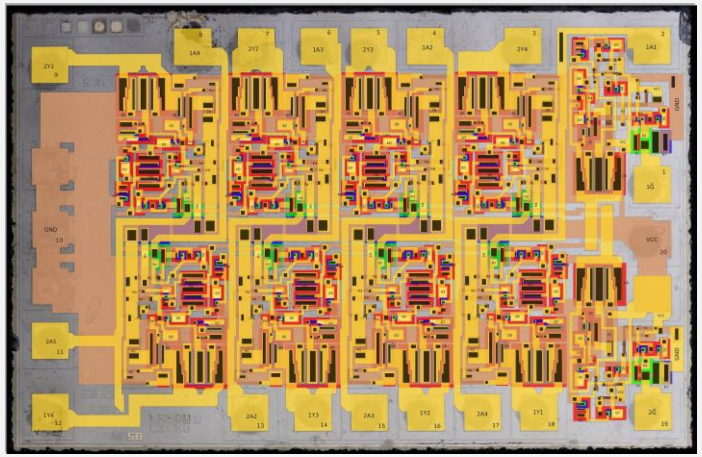

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Die photo of a Fairchild chip

Steps for Making ICs and Key Players

1

信越化學工業株式會社

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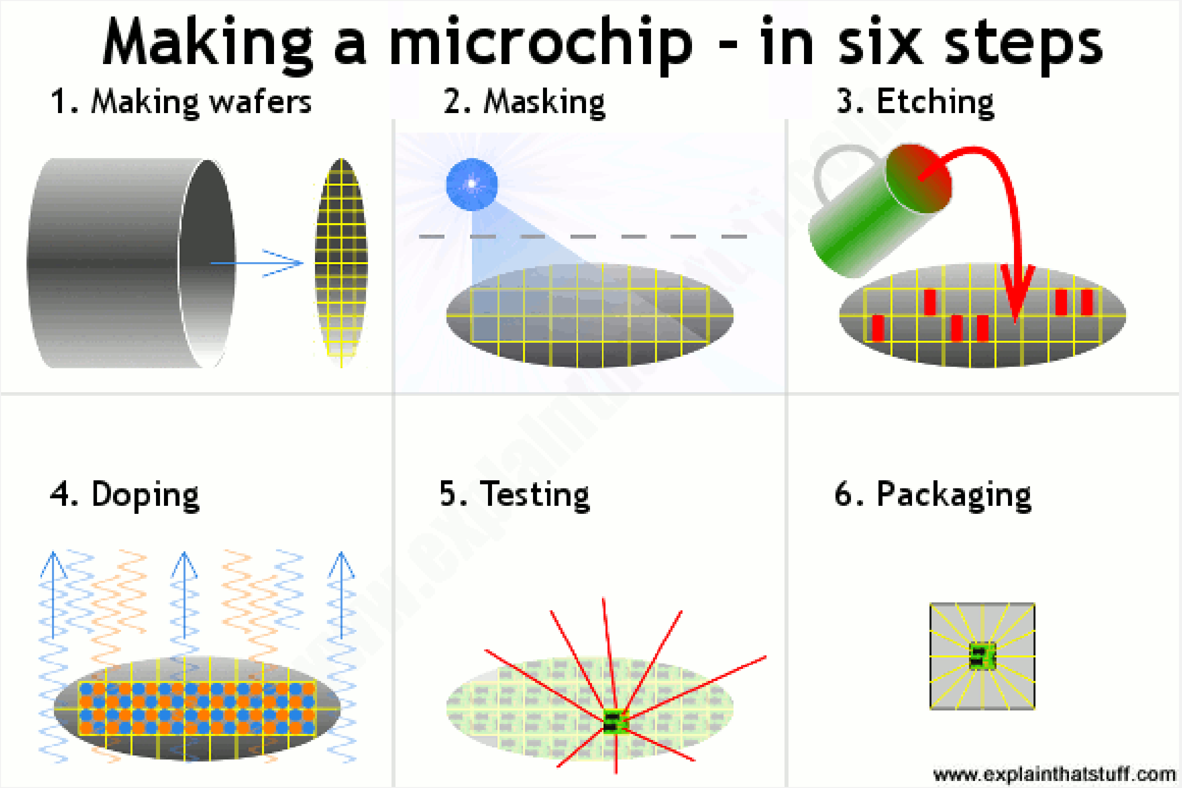

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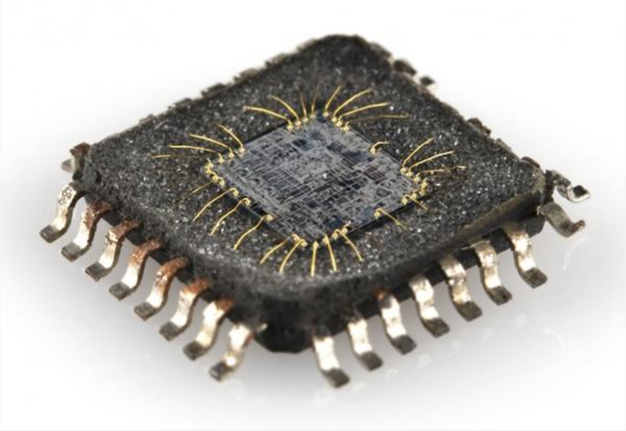
2~5

5~6

Anatomy of Integrated Circuits



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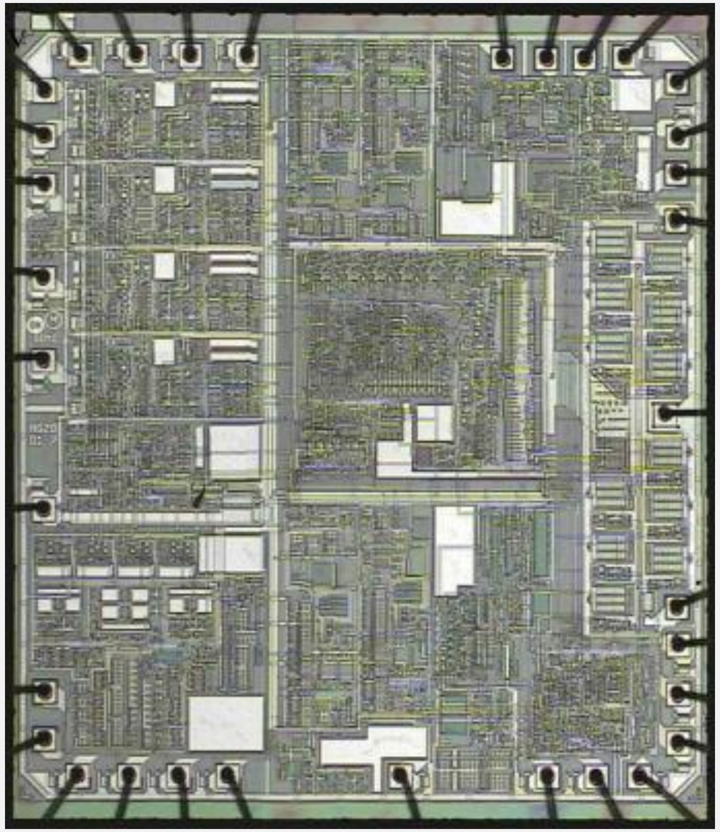
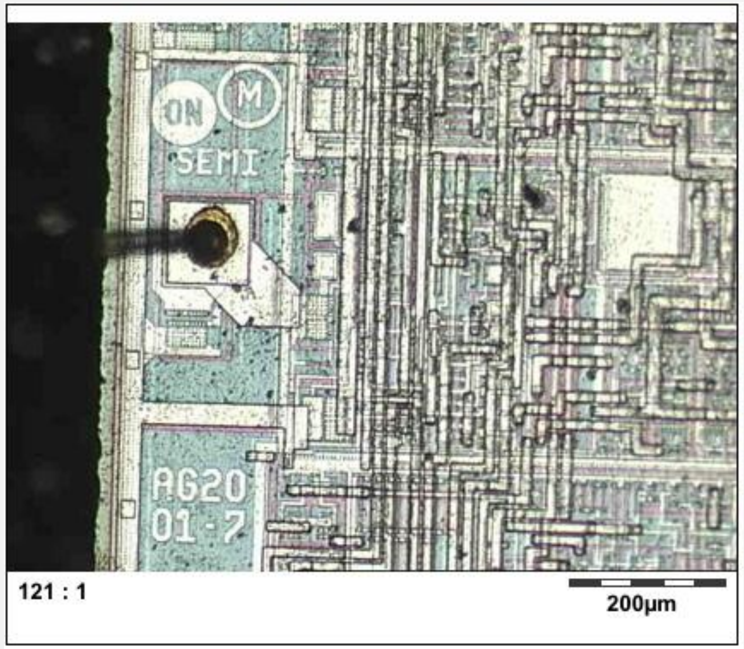




IC Design Eco-System

Design (Fabless) → Fabrication (Foundries) → Packaging

Example 1-1

- Today's microprocessors contain about 100 million transistors in a chip area of approximately 3cm x 3cm. (The chip is a few hundred microns thick.)
- Suppose integrated circuits were not invented and we attempted to build a processor using 100 million "discrete" transistors.
- If each device occupies a volume of 3mm x 3mm x 3mm, determine the minimum volume for the processor.
- What other issues would arise in such an implementation?

Solution:

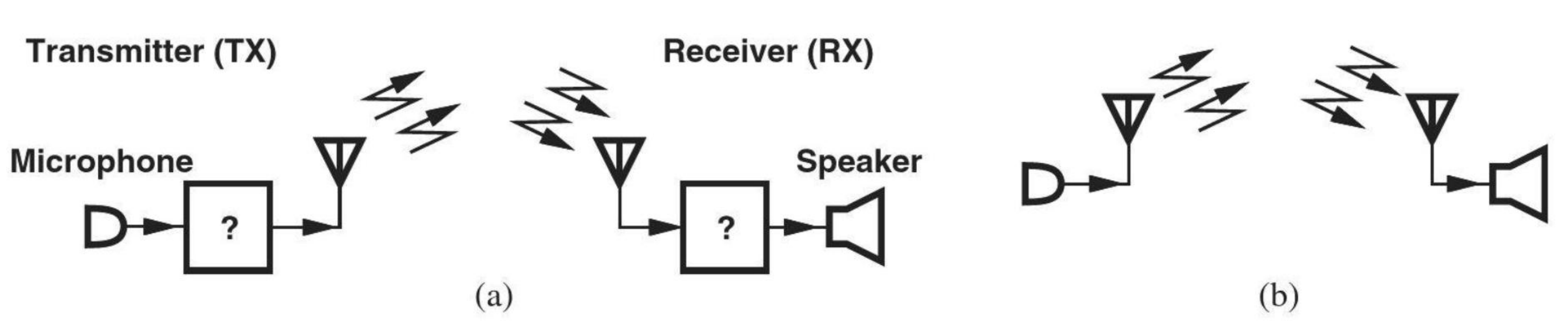
- The minimum volume is given by $27mm^3 \times 10^8$, i.e., a cube 1.4m on each side!
- Of course, the wires connecting the transistors would increase the volume substantially.
- In addition to occupying a large volume, this discrete processor would be extremely ***slow***; the signals would need to travel on wires as long as 1.4m!
- Furthermore, if each discrete transistor costs 1 cent and weighs 1g, each processor unit would be priced at one million dollars and weigh 100 tons!

Exercise

- How much power would such a system consume if each transistor dissipates 10μW?

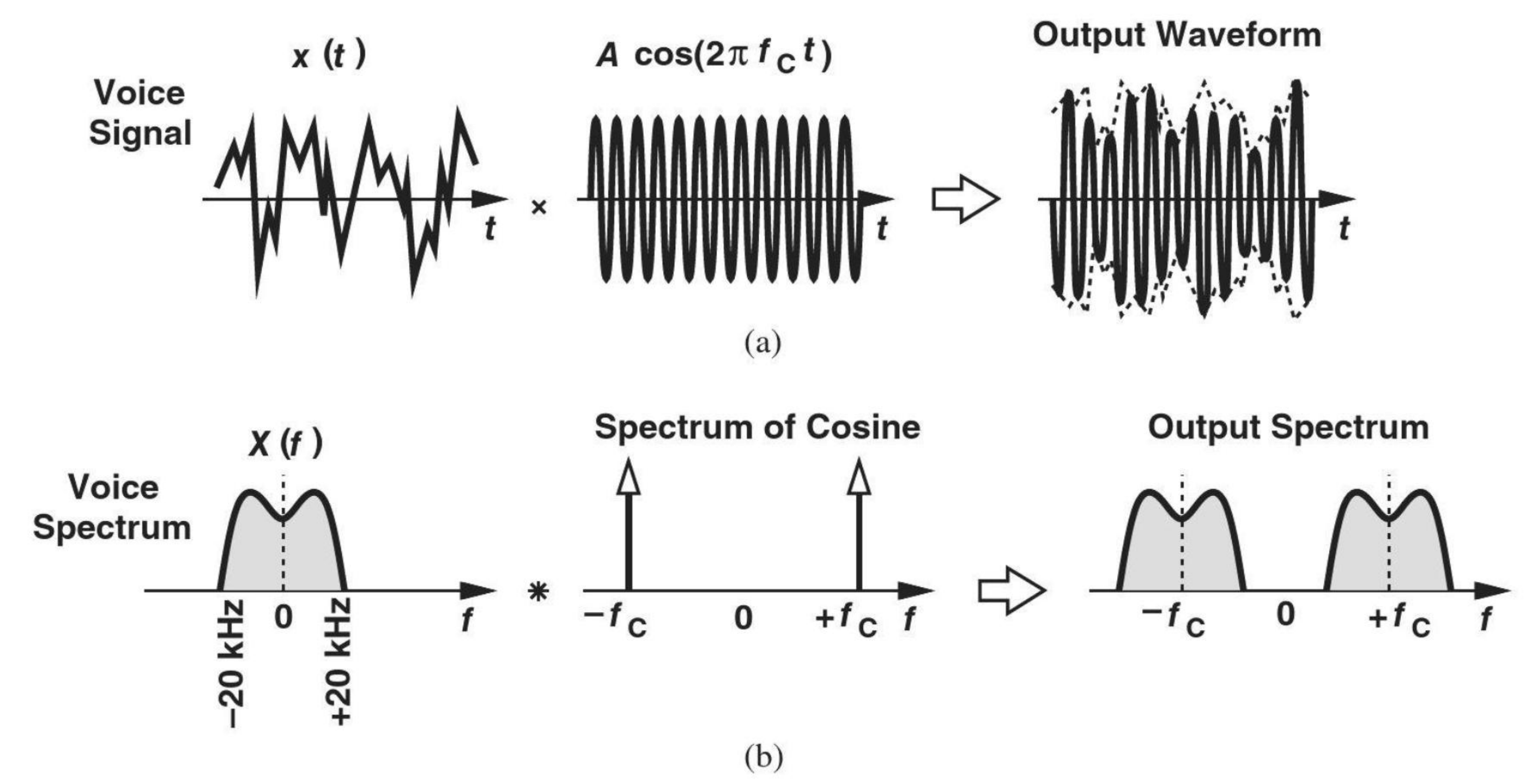
Examples of Electronic Systems: Cellular Telephone

- Cellular telephones were developed in the 1980s and rapidly became popular in the 1990s.
  - ✓ 1G(AMPS), 2G(GSM), 3G (CDMA), 4G(OFDM), 5G(Massive MIMO)
- Microelectronics exist in black boxes that process the received and transmitted voice signals.



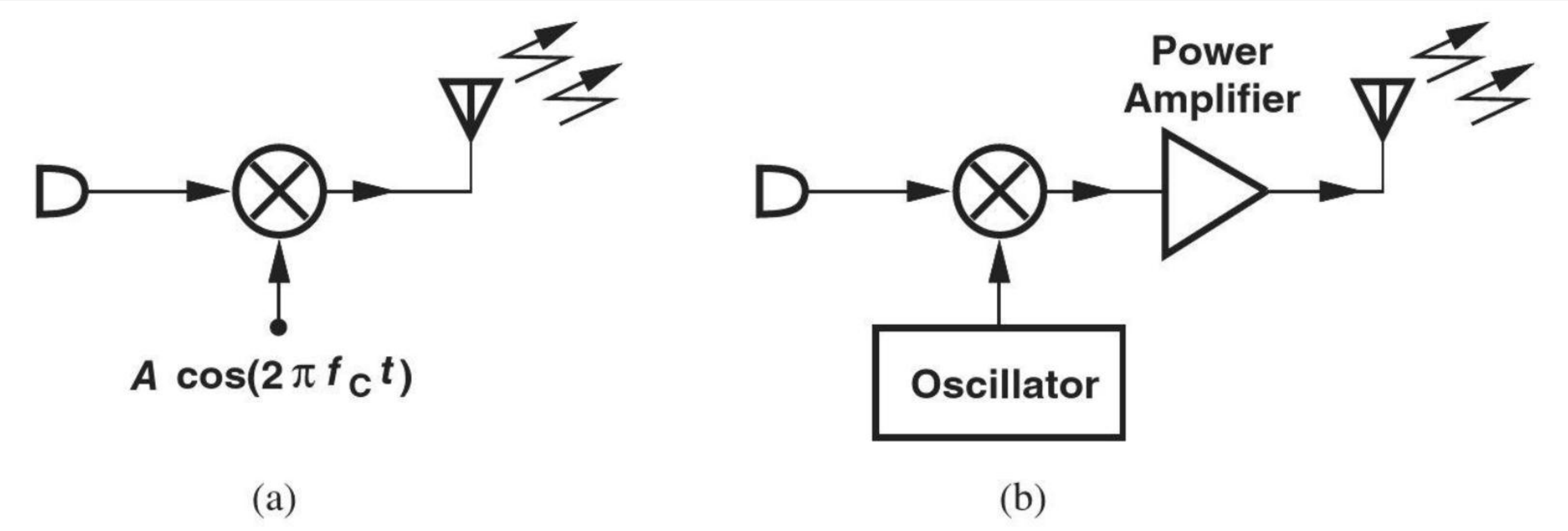
(a) Simplified view of a cellphone, (b) further simplification of transmit and receive paths.

Frequency Up-conversion



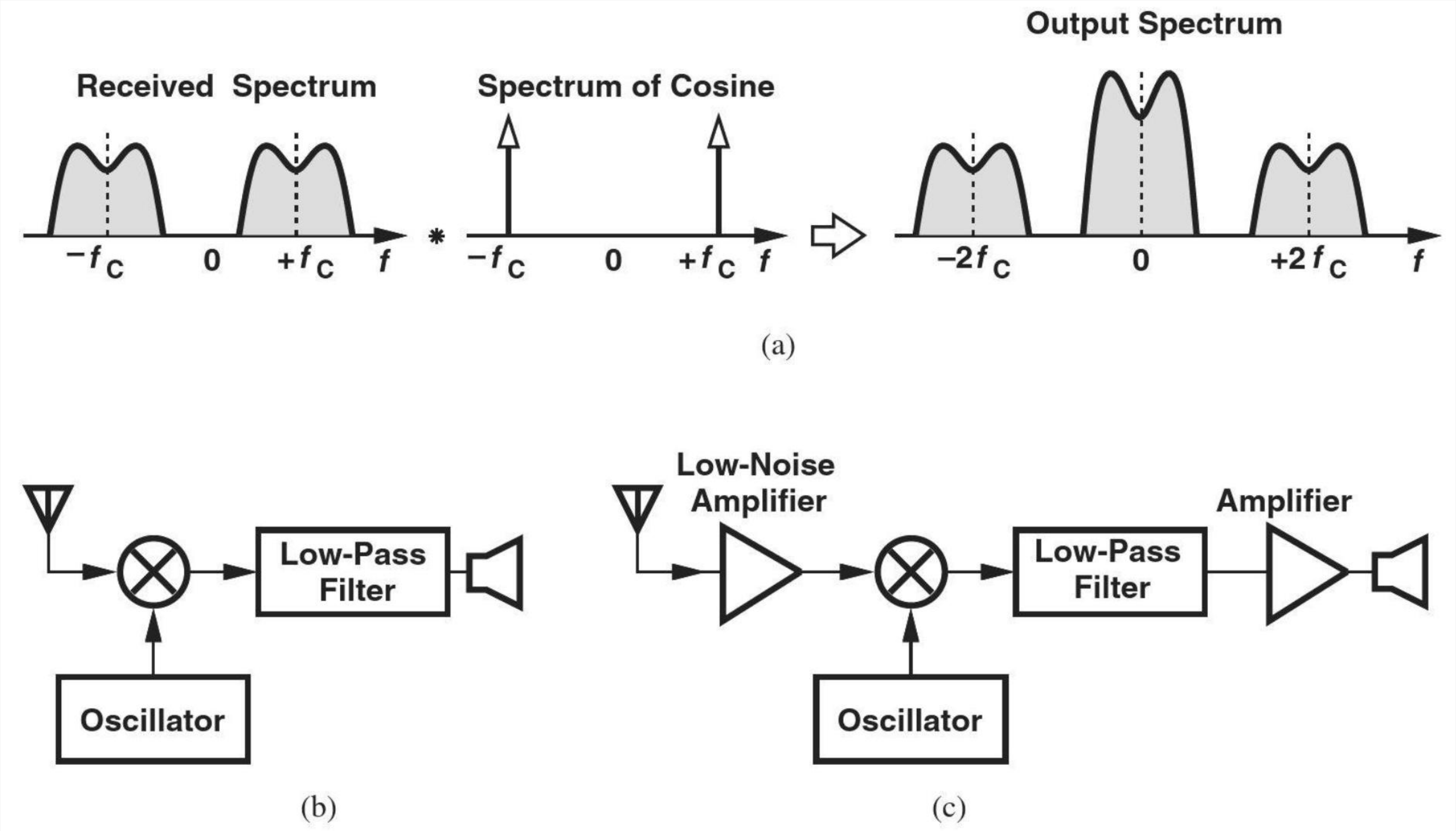
- Voice is "up-converted" by multiplying two sinusoids.
- When multiplying two sinusoids in time domain, their spectra are convolved in frequency domain.

Transmitter



- Two frequencies are multiplied and radiated by an antenna in (a).
- A power amplifier is added in (b) to boost the signal.

Receiver



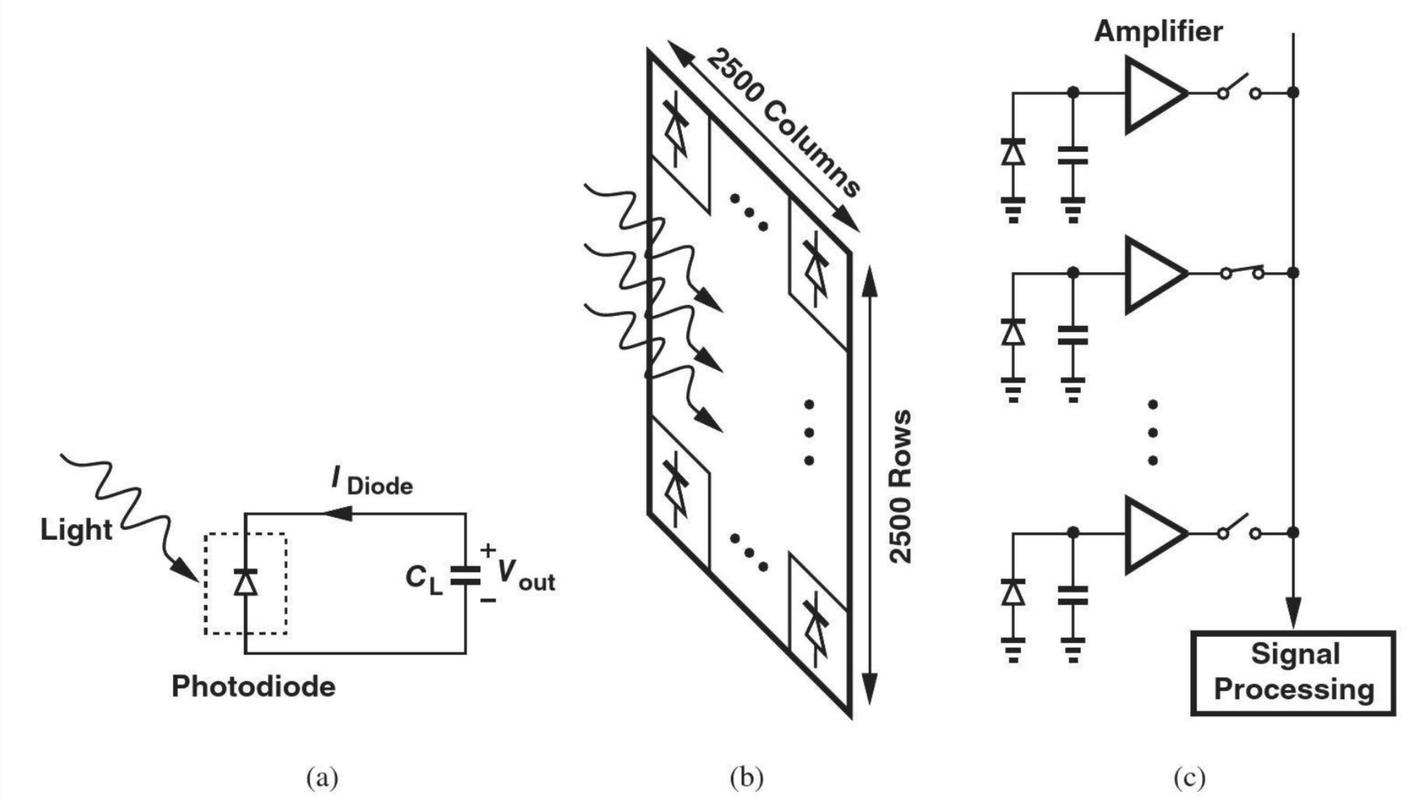
- High frequency is translated to DC by multiplying by f_C.
- A low-noise amplifier is needed for signal boosting without excessive noise.

Digital Camera

- With traditional cameras, we received no immediate feedback on the quality of the picture that was taken, we were very careful in selecting and shooting scenes to avoid wasting frames, we needed to carry bulky rolls of film, and we would obtain the final result only in printed form.
- With digital cameras, on the other hand, we have resolved these issues and enjoy many other features that only electronic processing can provide, e.g., transmission of pictures through cellphones or ability to retouch or alter pictures by computers.

Digital Camera

- The "front end" of the camera must convert light to electricity, a task performed by an array (matrix) of "pixels."



(a) Operation of a photodiode, (b) array of pixels in a digital camera, (c) one column of the array.

Digital Camera

- Each pixel consists of an electronic device (a "photodiode") that produces a current proportional to the intensity of the light that it receives.
- Now consider a camera with 6.25 million pixels (2500x2500) as in Fig. 1.5(b).
- How is the output voltage of each pixel sensed and processed? If each pixel contains its own electronic circuitry, the overall array occupies a very large area, raising the cost and the power dissipation considerably.
- We must therefore "time-share" the signal processing circuits among pixels.

Digital Camera

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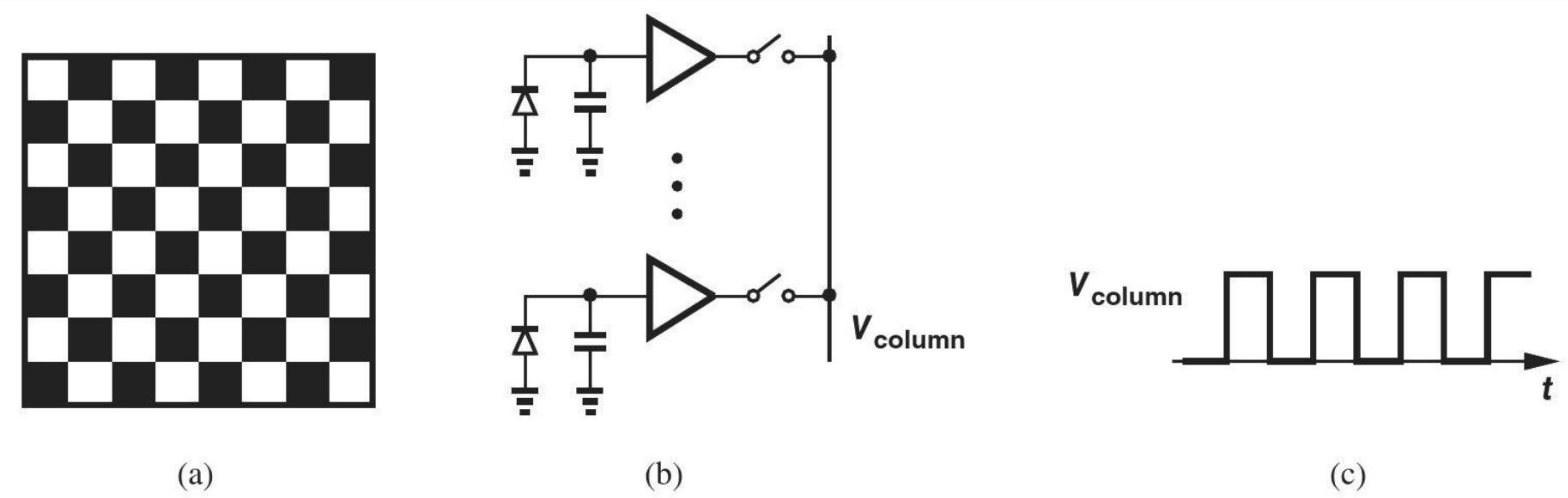
- We connect a wire to the outputs of all 2500 pixels in a “column,” turn on only one switch at a time.

Example 1-2

- A digital camera is focused on a chess board. Sketch the voltage produced by one column as a function of time.

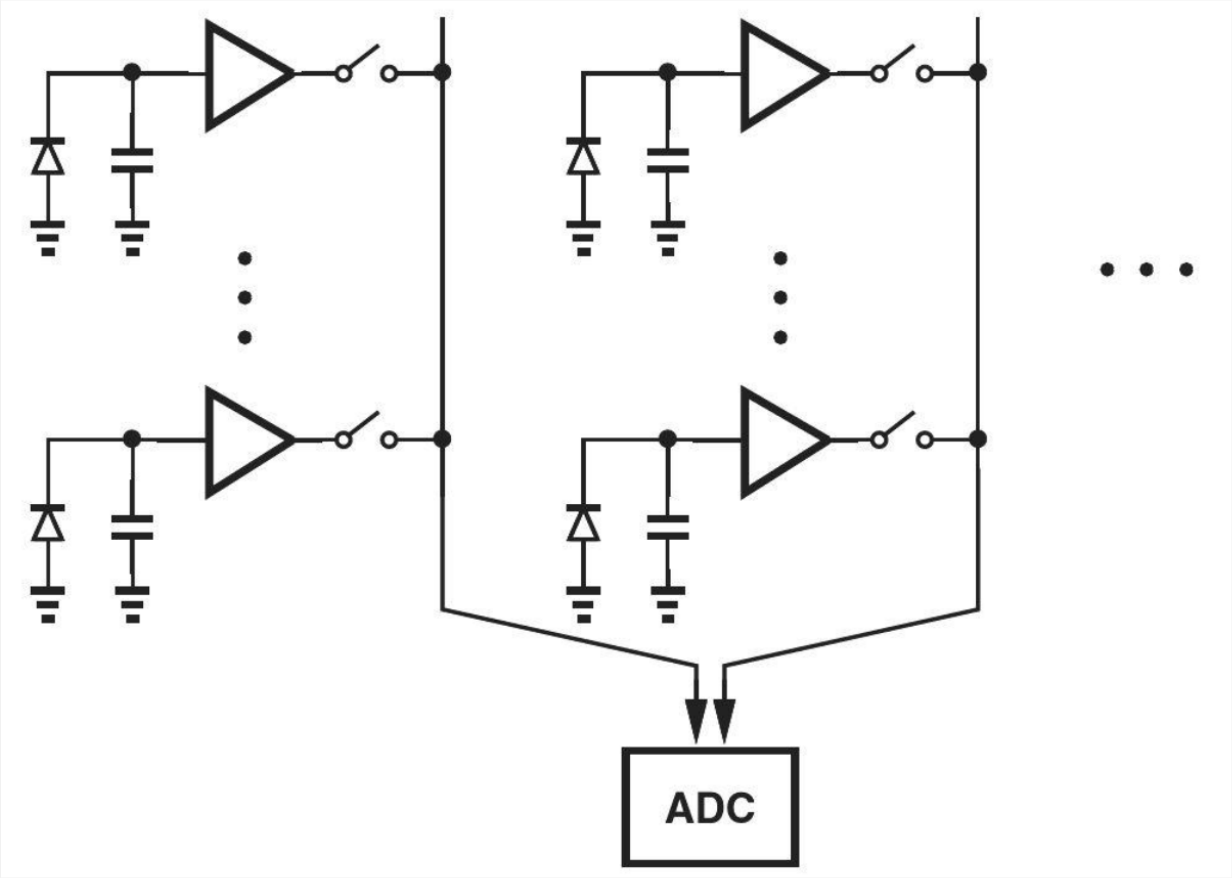


(a) Chess board captured by a digital camera, (b) sense circuits, and (c) voltage waveform on one column.

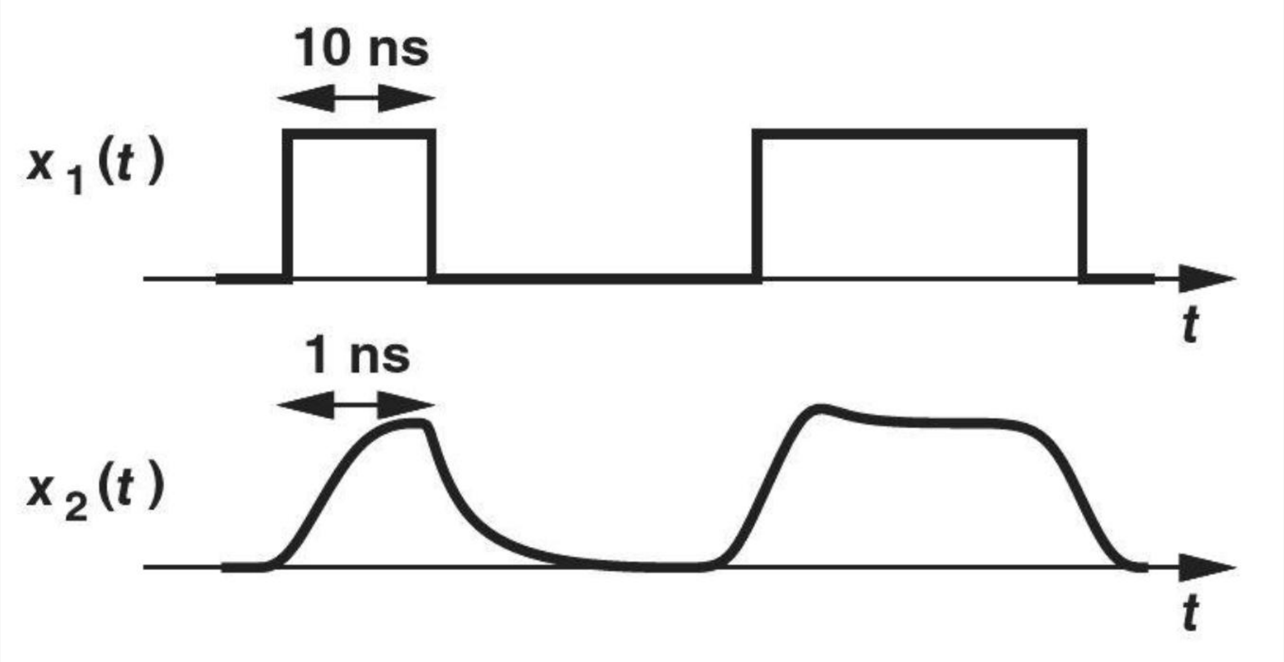
What does each signal processing block do?

- Since the voltage produced by each pixel is an analog signal and can assume all values within a range, we must first "digitize" it by means of an "analog-to-digital converter" (ADC).
- A 6.25 megapixel array must thus incorporate 2500 ADCs.
- Since ADCs are relatively complex circuits, we may time-share one ADC between every two columns as in next page (Fig. 1.7), but requiring that the ADC operate twice as fast (why?).
- In the extreme case, we may employ a single, very fast ADC for all 2500 columns.
- In practice, the optimum choice lies between these two extremes.

Sharing one ADC between two columns of a pixel array

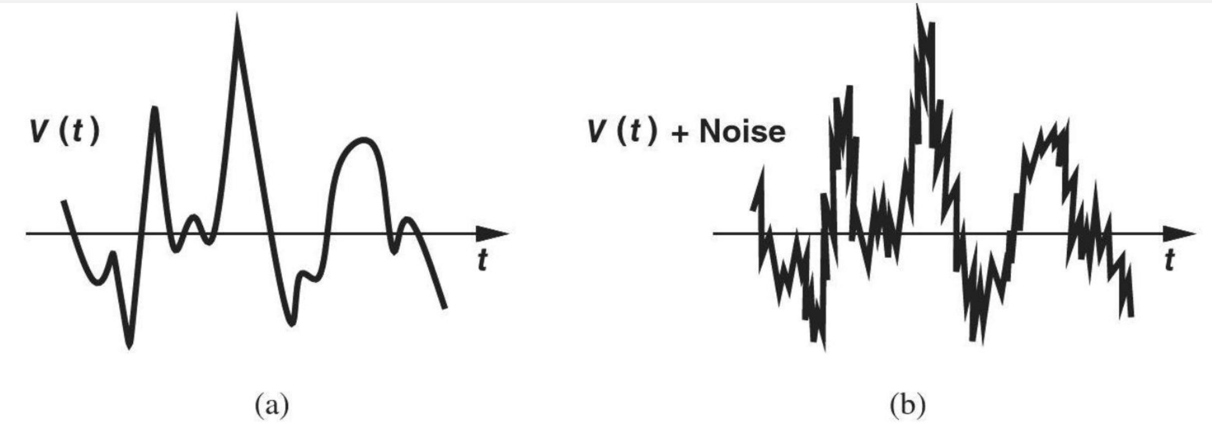


Digital or Analog?

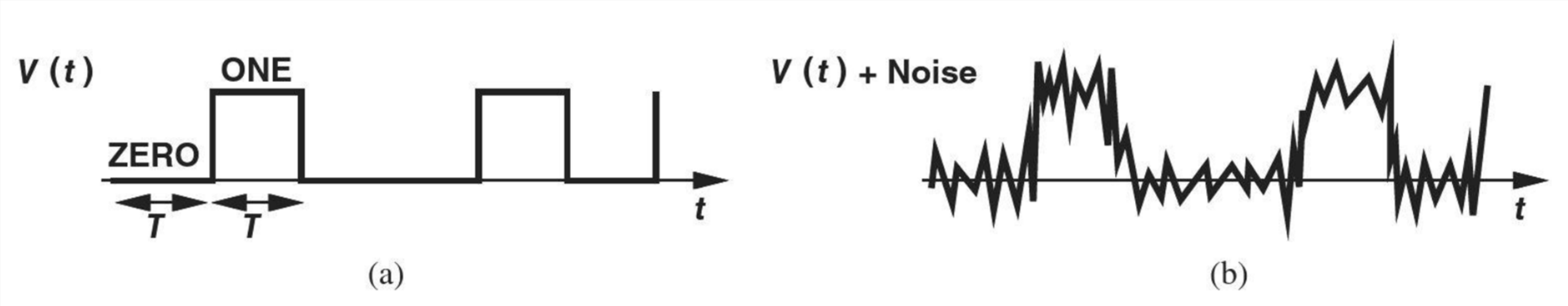


- x_1(t) is operating at 100Mb/s and x_2(t) is operating at 1Gb/s.
- A digital signal operating at very high frequency is very "analog".

Analog and Digital Signals

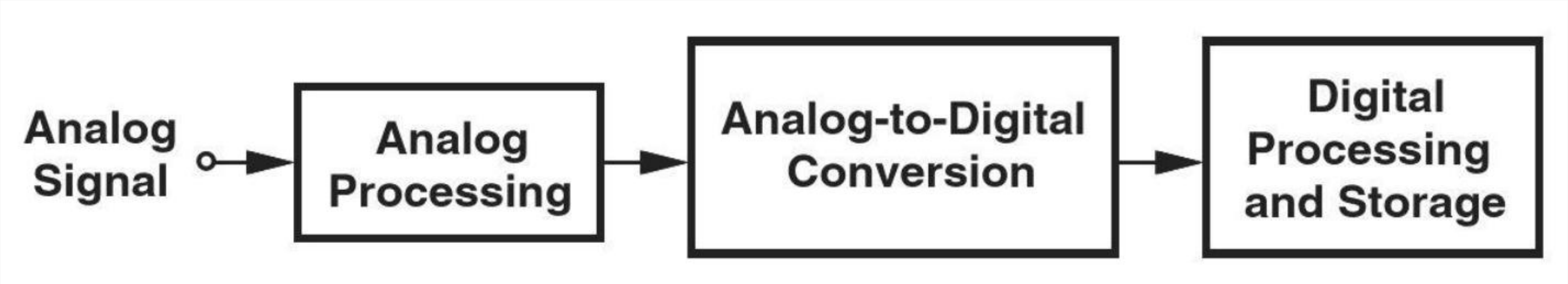


(a) Analog signal, (b) effect of noise on analog signal.

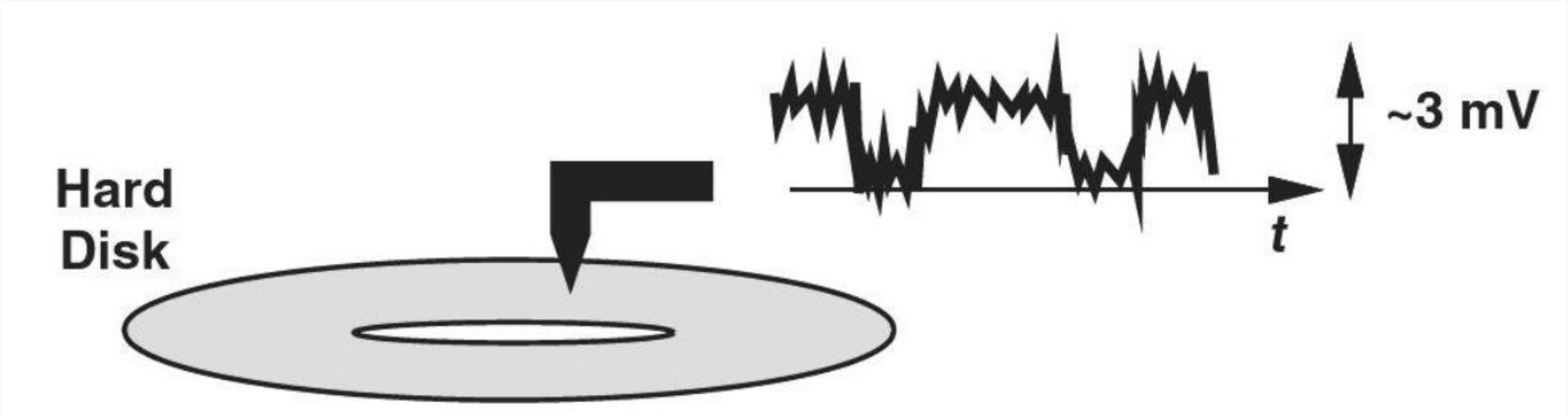


(a) Digital signal, (b) effect of noise on digital signal.

Signal processing in a typical system



Signal picked up from a hard disk in a computer



Example 1-3

- A cellphone receives a signal level of 20μV, but it must deliver a swing of 50mV to the speaker that reproduces the voice.
- Calculate the required voltage gain in decibels.

$$A_V = 20log\frac{50mV}{20uV}$$
$$\approx 68dB$$

- dB: The voltage gain of an amplifier is defined as $\frac{v_{out}}{v_{in}}$ and sometimes expressed in decibels (dB) as $20log\frac{v_{out}}{v_{in}}$.

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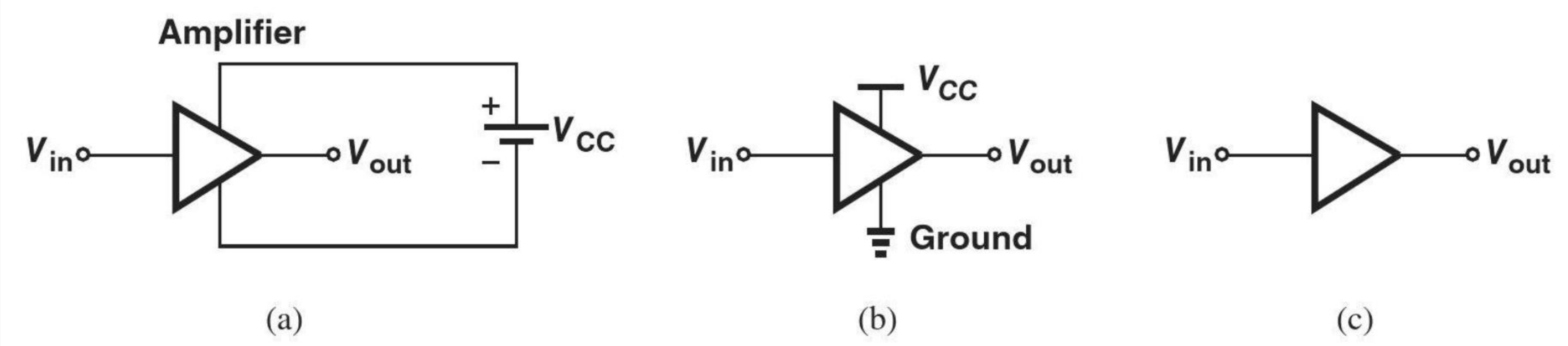
Exercise

- What is the output swing if the gain is 50 dB?

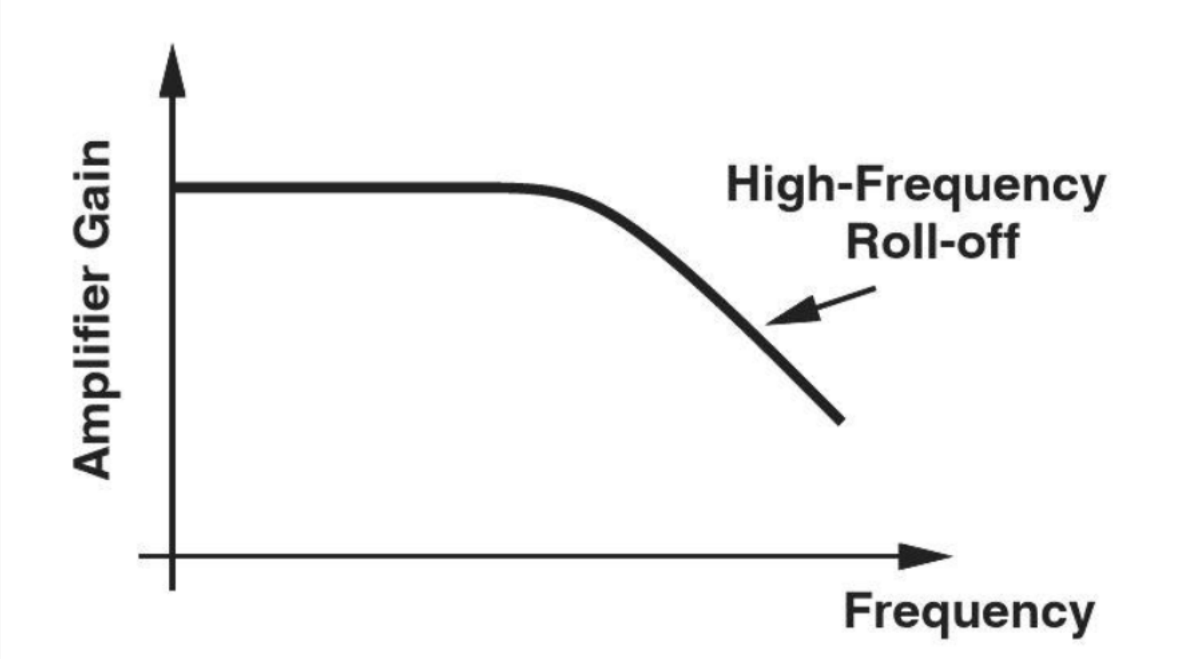
Analog Circuits



(a) General amplifier symbol along with its power supply,
(b) simplified diagram of (a),
(c) amplifier with supply rails omitted

Frequency Response

- What limits the speed of amplifiers?



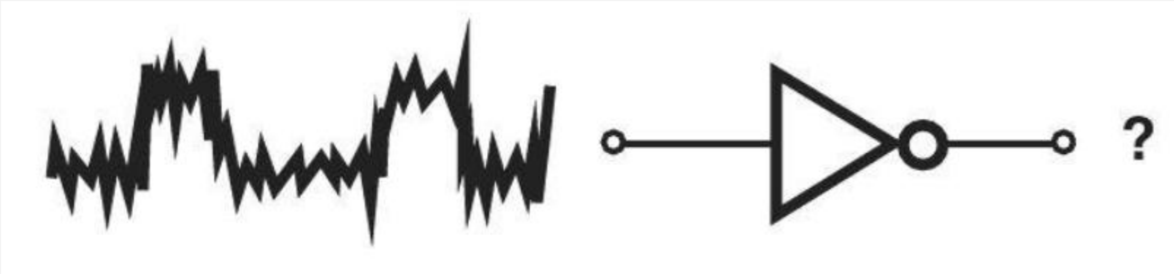
Roll-off of an amplifier's gain at high frequencies

Digital Circuits

- More than 80% of the microelectronics industry deals with digital circuits.
- Examples include microprocessors, static and dynamic memories, and digital signal processors.
- Recall from basic logic design that gates form "combinational" circuits, and latches and flipflops constitute "sequential" machines.
- The complexity, speed, and power dissipation of these building blocks play a central role in the overall system performance

Digital Logic

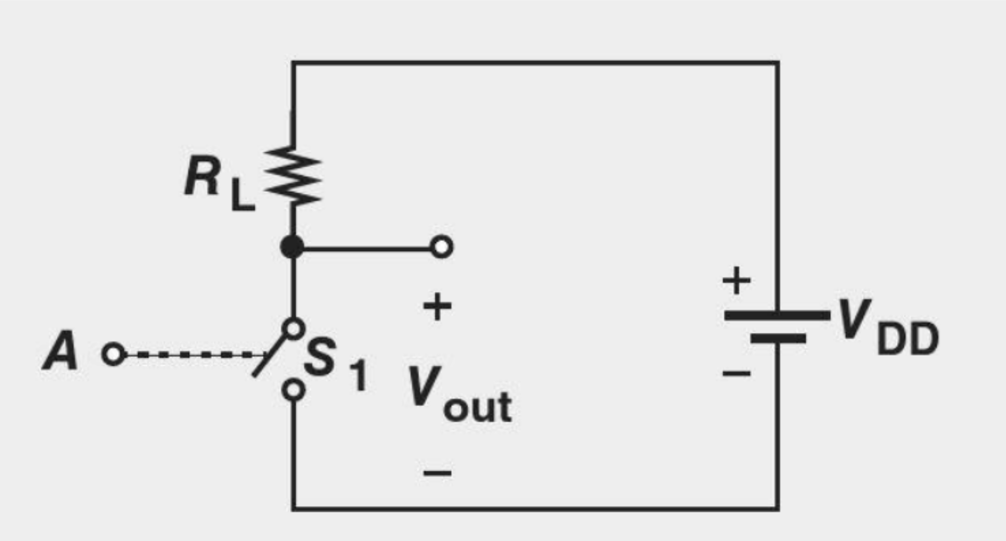
NOT and NOR gates



Response of a gate to a noisy input

Example 1-4

- Consider the circuit shown in Fig. 1.17, where switch is controlled by the digital input. That is, if A is high, S_1 is on and vice versa.
- Prove that the circuit provides the NOT function.

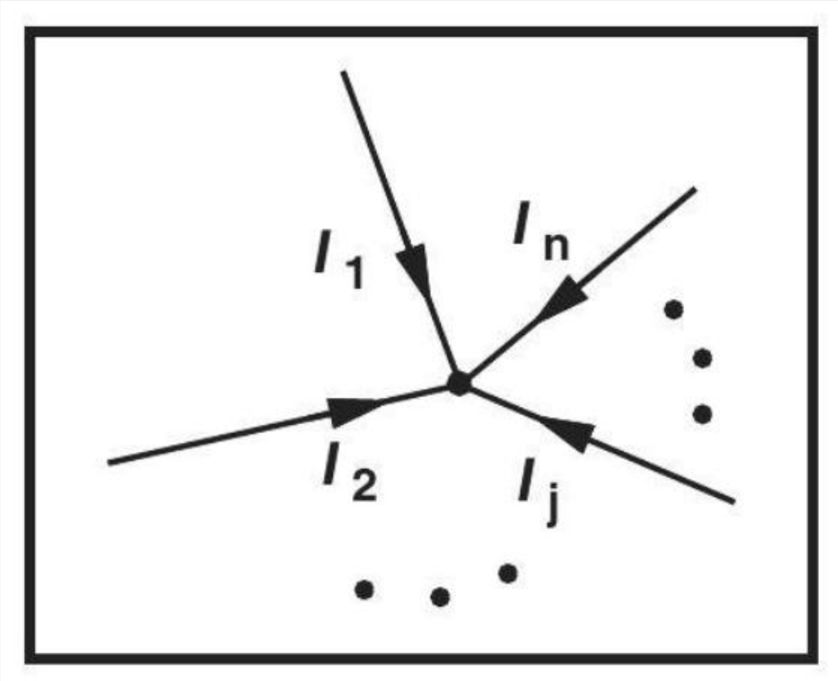


Solution:

- If A is high, S_1 is on, forcing V_{out} to zero.
- On the other hand, if A is low, S_1 remains off, drawing no current from R_L.
- As a result, the voltage drop across R_L is zero and hence V_{out} = V_{DD}; i.e., the output is high.
- We thus observe that, for both logical states at the input, the output assumes the opposite state.

Basic Circuit Theorems

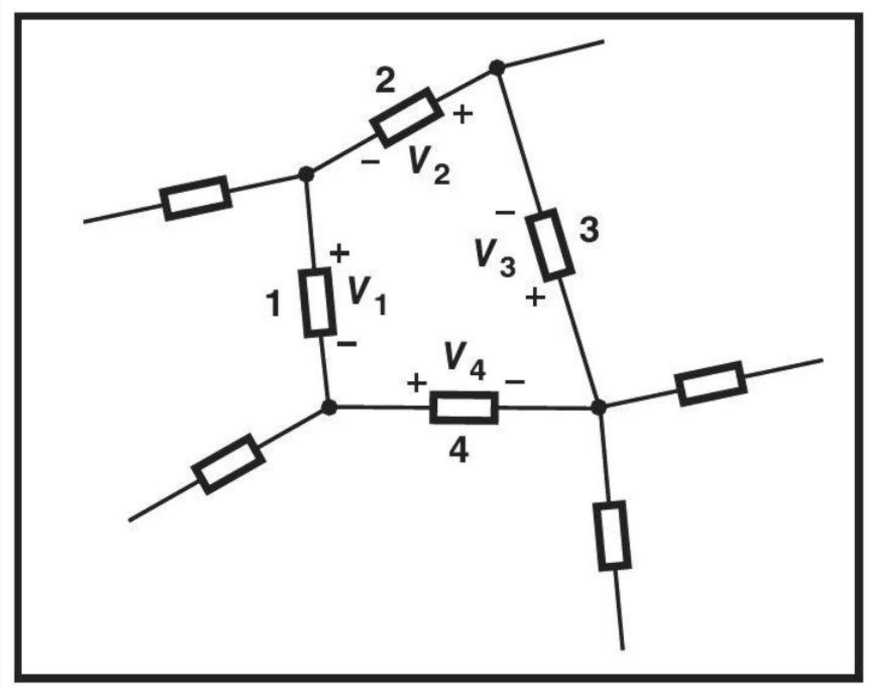
- KCL: Kirchoff's current law states that the sum of all currents flowing into any node is zero.



$$\sum_j I_j = 0$$

Basic Circuit Theorems

- KVL: Kirchoff's voltage law states that the sum of all voltages around any loop is zero.



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References

[1] https://en.wikipedia.org/wiki/Vacuum_tube

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